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## Afferent neuron circuit and mechanoreceptive system

### Abstract

Disclosed is an afferent neuron circuit, which includes: a resistance Rc and a volatile threshold switching device TS, wherein the volatile threshold switching device TS is provided with a parasitic capacitor Cparasitic; a first end of the resistance Rc serves as a signal input terminal, and a second end of the resistance Rc serves as a signal output terminal; and a first end of the volatile threshold switching device TS is connected to the signal output terminal, and a second end of the volatile threshold switching device TS is grounded. The afferent neuron circuit provided in the content of the present disclosure has a simple structure and good scalability and is suitable for large-scale integration.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims the priority of Chinese Patent Application No. 201911127991.9 filed before the Chinese National Intellectual Property Administration on Nov. 18, 2019, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to an afferent neuron circuit and a mechanoreceptive system, which belong to the technical field of brain-like bionics.

BACKGROUND OF THE INVENTION

(3) The present disclosure provides an afferent neuron circuit and a mechanoreceptive system. An artificial afferent neuron is a component that converts an external stimulation signal into a systematic impulse signal and plays an important role in the construction of an impulse neural network system. At present, an afferent neuron circuit is mainly based on CMOS circuits, so that it has a complex structure and poor scalability and thereby is not applicable to large-scale integration use.

SUMMARY OF THE INVENTION

(4) The present disclosure provides an afferent neuron circuit and a mechanoreceptive system, which solves the technical problem in the existing technology that afferent neuron circuits have a complex structure and poor scalability and thereby are not applicable to large-scale integration use.

(5) In one aspect of the present disclosure, provided is an afferent neuron circuit, which includes a resistance Rc and a volatile threshold switching device TS, wherein the volatile threshold switching device TS has a parasitic capacitor Cparasitic; a first end of the resistance Rc is used as a signal input terminal, and a second end of the resistance Rc is used as a signal output terminal; a first end of the volatile threshold switching device TS is connected to the signal output terminal, and a second end of the volatile threshold switching device TS is grounded.

(6) In some embodiments, the afferent neuron circuit adopts an integrated semiconductor structure.

(7) In some embodiments, the volatile threshold switching device TS includes: a first substrate **11**; a first isolation layer **12** formed on the first substrate **11**; a first lower electrode **13** formed on the first isolation layer **12**; a second isolation layer **14** formed on the first lower electrode **13**; a first functional layer **15** formed on the second isolation layer **14**, wherein a middle part of the first functional layer **15** is connected to the first lower electrode **13** through an etched via hole; and a first intermediate electrode **16** formed on the first functional layer **15**, wherein a first resistive film **17** is deposited on the first intermediate electrode **16** to serve as the resistance Rc, and a first upper electrode **18** is deposited on the first resistive film **17**.

(8) In some embodiments, the first substrate **11** is a silicon wafer; the first isolation layer **12** is a SiO.sub.2 layer; a material of the first lower electrode **13** is TiN, Poly-Si, Pd, Pt, W, or Au; the second isolation layer **14** is a SiO.sub.2 layer; a material of the first functional layer **15** is NbO.sub.2, VO.sub.2, SiTe, SiO.sub.2:Ag, a-Si:Cu, a-Si:Ag, or AM.sub.4Q.sub.8; a material of the first intermediate electrode **16** is TiN, Poly-Si, Pd, Pt, W, Cu, Ag or Au; a resistance value of the first resistive film **17** is greater than 5 times of a resistance value of the first functional layer **15** in a low-resistance state and is less than ⅕ of a resistance value of the first functional layer **15** in a high-resistance state; wherein the AM.sub.4Q.sub.8 is a mixed material of Ga, Ge, V, Nb, Ta, Mo, S, and Se.

(9) In some embodiments, the volatile threshold switching device TS includes: a second substrate **21**; a third isolation layer **22** formed on the second substrate **21**; a second lower electrode **23** formed on the third isolation layer **22**; a second functional layer **24** formed on the third isolation layer **23**; and a second intermediate electrode **25** formed on the second functional layer **24**, wherein a second resistive film **26** is deposited on the second intermediate electrode **25** to serve as the resistance Rc, and a second upper electrode **27** is deposited on the second resistive film **26**.

(10) In some embodiments, the second substrate **21** is a silicon wafer; the third isolation layer **22** is a SiO.sub.2 layer; a material of the second lower electrode **23** is TiN, Poly-Si, Pd, Pt, W or Au; a material of the second functional layer **24** is NbO.sub.2, VO.sub.2, SiTe, SiO.sub.2:Ag, a-Si:Cu, a-Si:Ag, or AM.sub.4Q.sub.8; a material of the second intermediate electrode **25** is TiN, Poly-Si, Pd, Pt, W, Cu, Ag, or Au; a resistance value of the second resistive film **26** is greater than 5 times of a resistance value of the second functional layer **24** in a low-resistance state and is less than ⅕ of a resistance value of the first functional layer **15** in a high-resistance state; and wherein said the AM.sub.4Q.sub.8 is a mixed material of Ga, Ge, V, Nb, Ta, Mo, S, and Se.

(11) In another aspect of the present disclosure, provided is a mechanoreceptive system, which includes: the aforementioned afferent neuron circuit; and, a first piezoelectric film **19** formed on

the first upper electrode **18** of the aforementioned afferent neuron circuit, and a first ground electrode **10** deposited on the first piezoelectric film **19**.
(12) In some embodiments, a material of the first piezoelectric film **19** may be BaTiO.sub.3/bacterial cellulose, K.sub.0.5Na.sub.0.5NbO.sub.3—BaTiO.sub.3/polyvinylidene fluoride, crystal, lithium gallate, lithium germanate, titanium germanate, or lithium tantalate.
(13) In yet another aspect of the present disclosure, provided is a mechanoreceptive system, which includes: the aforementioned afferent neuron circuit; and, a second piezoelectric film **28** formed on the second upper electrode **27** of the aforementioned afferent neuron circuit, and a second ground electrode **29** deposited on the second piezoelectric film **28**.
(14) In some embodiments, a material of the second piezoelectric film **28** may be BaTiO.sub.3/bacterial cellulose, K.sub.0.5Na.sub.0.5NbO.sub.3—BaTiO.sub.3/polyvinylidene fluoride, crystal, lithium gallate, lithium germanate, titanium germanate, or lithium tantalate.
(15) The present disclosure provides an afferent neuron circuit and a mechanoreceptive system, wherein the afferent neuron circuit with a simple structure is formed by connecting a resistance Rc and a volatile threshold switching device TS in series. A connection point between the volatile threshold switching device TS and the resistance Rc is used as a signal output terminal. An input signal is a voltage signal or a current signal, and an output signal is an oscillation frequency signal. An oscillation frequency of the output signal is related to the strength of the input signal. The volatile threshold switching device TS is provided with a parasitic capacitor C.sub.parasitic, so that when there is an input signal at the signal input terminal, the TS device is charged by the circuit through the resistance Rc; when a voltage across the parasitic capacitor C.sub.parasitic exceeds a switching voltage of the TS device, the TS device switches to a low-resistance state, and the capacitor discharges through the TS device; when the voltage across the parasitic capacitor Cparasitic drops to a holding voltage of the TS device, the TS device switches to a high-resistance state, and the parasitic capacitor Cparasitic is charged again through the resistance Rc. These are repeated over and over again to form an oscillation output pulse signal, thereby achieving the so-called functional integrity of an afferent neuron.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) By reading the detailed description of the preferred embodiments below, various other advantages and benefits will become clear to those of ordinary skill in the art. The drawings are only used for the purpose of illustrating the preferred embodiments and should not be considered as a limitation to the present disclosure. Also, throughout the drawings, the same reference symbol is configured to denote the same component.
(2) FIG. **1** is a schematic principle diagram of an afferent neuron circuit provided by the Embodiment 1 of the disclosure;
(3) FIG. **2** is a schematic principle diagram of a mechanoreceptive system provided by the Embodiment 1 of the disclosure;
(4) FIG. **3** is a schematic diagram of an integrated semiconductor structure of the afferent neuron circuit provided by the Embodiment 1 of the disclosure;
(5) FIG. **4** is a schematic diagram of processing flow of the integrated semiconductor structure of the afferent neuron circuit provided by the Embodiment 1 of the disclosure;
(6) FIG. **5** is a schematic diagram of an integrated semiconductor structure of the mechanoreceptive system provided by the Embodiment 1 of the disclosure;
(7) FIG. **6** is a schematic diagram of processing flow of the integrated semiconductor structure of the mechanoreceptive system provided by the Embodiment 1 of the disclosure;
(8) FIG. **7** is a schematic diagram of an integrated semiconductor structure of an afferent neuron

circuit provided by the Embodiment 2 of the disclosure;
(9) FIG. **8** is a schematic diagram of the processing flow of the integrated semiconductor structure of the afferent neuron circuit provided by the Embodiment 2 of the disclosure;
(10) FIG. **9** is a schematic diagram of an integrated semiconductor structure of a mechanoreceptive system provided by the Embodiment 2 of the disclosure;
(11) FIG. **10** is a schematic diagram of the processing flow of the integrated semiconductor structure of the mechanoreceptive system provided by the Embodiment 2 of the disclosure;
(12) FIG. **11** shows I-V characteristic curves of the afferent neuron circuits provided by the Embodiment 1 and the Embodiment 2 of the disclosure;
(13) FIG. **12** shows output oscillation curves of the afferent neuron circuits provided by the Embodiment 1 and the Embodiment 2 of the disclosure that are subjected to two input voltages;
(14) FIG. **13** shows a relationship between an output frequency and an input voltage of the afferent neuron circuits provided by the Embodiment 1 and the Embodiment 2 of the disclosure; and
(15) FIG. **14** shows an output voltage curve of a piezoelectric film when pressure is exerted, an oscillation output curve, and an oscillation output frequency curve of the afferent neuron circuits provided by the Embodiment 1 and the Embodiment 2 of the disclosure.

DETAILED DESCRIPTION OF THE INVENTION

(16) Hereinafter, exemplary embodiments of the present disclosure will be described in more detail with reference to the accompanying drawings. Although the drawings show exemplary embodiments of the present disclosure, it should be understood that the present disclosure can be implemented in various forms and should not be limited by the embodiments set forth herein. Rather, these embodiments are provided to enable a more thorough understanding of the present disclosure and to fully convey the scope of the present disclosure to those skilled in the art.
(17) The technical solution of the present disclosure will be further described in detail below through the accompanying drawings and specific embodiments.

Embodiment 1 of the Present Disclosure

(18) Referring to FIG. **1**, the present embodiment provides an afferent neuron circuit, including a resistance Rc and a volatile threshold switching device TS. The volatile threshold switching device TS is provided with a parasitic capacitor Cparasitic. A first end of the resistance Rc serves as a signal input terminal, and a second end of the resistance Rc serves as a signal output terminal. A first end of the volatile threshold switching device TS is connected to the signal output terminal, and a second end of the volatile threshold switching device TS is grounded.
(19) An input signal is a voltage signal or a current signal, and an output signal is an oscillation frequency signal. An oscillation frequency of the output signal is related to the strength of the input signal.
(20) When there is an input signal at the signal input terminal, the parasitic capacitor Cparasitic is charged by the circuit through the resistance Rc.
(21) In the case that a voltage across the parasitic capacitor Cparasitic exceeds a switching voltage of the volatile threshold switching device TS, the volatile threshold switching device TS switches to a low-resistance state, and the parasitic capacitor Cparasitic discharges through the volatile threshold switching device TS.
(22) In the case that the voltage across the parasitic capacitor Cparasitic drops to a holding voltage of the volatile threshold switching device TS, the volatile threshold switching device TS switches to a high-resistance state, and the parasitic capacitor Cparasitic is charged again through the resistance Rc. The charge and discharge processes are repeated over and over again to form an oscillation output pulse signal.
(23) Referring to FIG. **2**, on the basis of the afferent neuron circuit, a piezoelectric component is attached so as to form a mechanoreceptive system. In this case, the piezoelectric component is configured to convert a sensed mechanical stimulus into a related voltage signal. In a resting state, the piezoelectric component has no output; that is, its static power consumption is zero. The

mechanoreceptive system converts a voltage signal generated by the piezoelectric component into a relevant oscillation frequency output signal during operation. An oscillation frequency of the output signal is related to a mechanical strength sensed by the piezoelectric component.
(24) Referring to FIG. **3**, in this embodiment, the afferent neuron circuit uses an integrated semiconductor structure in order to improve structural scalability and facilitate integration use of the afferent neuron circuit.
(25) Specifically, the volatile threshold switching device TS includes: a first substrate **11**; a first isolation layer **12**, which is formed on the first substrate **11**; a first lower electrode **13**, which is formed on the first isolation layer **12**; a second isolation layer **14**, which is formed on the first lower electrode **13**; a first functional layer **15**, which is formed on the second isolation layer **14**, wherein a middle part of the first functional layer **15** is connected to the first lower electrode **13** through an etched via hole; and a first intermediate electrode **16**, which is formed on the first functional layer **15**; wherein a first resistive film **17** is deposited on the first intermediate electrode **16** to serve as the resistance Rc, and a first upper electrode **18** is deposited on the first resistive film **17**.
(26) In the present embodiment, the first substrate **11** is a silicon wafer; the first isolation layer **12** is a SiO.sub.2 layer; the material of the first lower electrode **13** is TiN, Poly-Si, Pd, Pt, W, or Au, and may also be selected from other inert conductive materials; the second isolation layer **14** is a SiO.sub.2 layer; the material of the first functional layer **15** is NbO.sub.2, VO.sub.2, SiTe, SiO.sub.2:Ag, a-Si:Cu, a-Si:Ag or AM.sub.4Q.sub.8; the material of the functional layer may, but be not limited to, be selected from the above materials, and any material having a volatile threshold switching characteristic can be used; the material of the first intermediate electrode **16** is TiN, Poly-Si, Pd, Pt, W, Cu, Ag or Au; the resistance value of the first resistive film **17** is greater than 5 times of a resistance value of the first functional layer **15** in a low-resistance state and is less than ⅕ of a resistance value of the first functional layer **15** in a high-resistance state; and in this case, the AM.sub.4Q.sub.8 is a mixed material of Ga, Ge, V, Nb, Ta, Mo, S, and Se.
(27) Referring to FIG. **4**, a process for preparing an afferent neuron circuit and a mechanoreceptive system will be described in detail below.
(28) At Step **1**, a silicon wafer serves as a first substrate **11**, and oxidation is applied on the silicon wafer to form a SiO.sub.2 layer as a first isolation layer **12**. The thickness of the SiO.sub.2 layer is 100 nm˜300 nm, and the thickness of the SiO.sub.2 layer may also be decreased or increased according to an actual process condition.
(29) At Step **2**, a first lower electrode **13** is deposited on the SiO.sub.2 layer. The thickness of the first lower electrode **13** is 10 nm˜100 nm.
(30) At Step **3**, a second isolation layer **14** is deposited on the first lower electrode **13**. The second isolation layer **14** is a SiO.sub.2 layer with a thickness of 100 nm to 300 nm. The thickness of the SiO.sub.2 layer may also be decreased or increased according to an actual process condition.
(31) At Step **4**, by using an etching process, a hole is etched on the deposited films prepared in Steps **1** to **3**. The depth of the hole directly reaches inside of the first isolation layer **12**, and the first isolation layer **12** is etched by a certain depth to ensure that the cross-section of the first lower electrode **13** is completely exposed. The first isolation layer **12** should also retain a sufficient thickness after etching.
(32) At Step **5**, a first functional layer **15** is deposited on the etched hole. The thickness of the first functional layer **15** is 5 nm˜50 nm.
(33) At Step **6**, a first intermediate electrode **16** is deposited on the first functional layer **15**. The thickness of the first intermediate electrode **16** is 10 nm˜100 nm.
(34) At Step **7**, a first resistive film **17** is deposited on the first intermediate electrode **16** to serve as a resistance Rc. The resistance value of the first resistive film **17** is 1 Ω˜1 MΩ, which depends on a high-resistance or low-resistance state of the functional layer that is actually deposited.
(35) At Step **8**, a first upper electrode **18** is deposited on the first resistive film. A material of the

first upper electrode **18** is not limited herein.

(36) Referring to FIG. **5**, the present disclosure further provides a mechanoreceptive system, including:

(37) the afferent neuron circuit;

(38) a first piezoelectric film **19**, which is formed on the first upper electrode **18**; and

(39) a first ground electrode **10**, which is deposited on the first piezoelectric film **19**.

(40) In some schemes, a material of the first piezoelectric film **19** may be BaTiO.sub.3/bacterial cellulose, K.sub.0.5Na.sub.0.5NbO.sub.3—BaTiO.sub.3/polyvinylidene fluoride, crystal, lithium gallate, lithium germanate, titanium germanate, or lithium tantalate. The piezoelectric material may, but be not limited to, be selected from the above materials, and any material having a piezoelectric characteristic can be applied. However, the thermal budget of the functional layers should be taken into account during integration so as to determine the annealing temperature of the piezoelectric material.

(41) Referring to FIG. **6**, a method for preparing a mechanoreceptive system is provided according to the disclosure on the basis of the method mentioned above and includes steps **9** and **10**.

(42) At Step **9**, on the basis of the aforementioned afferent neuron circuit, a first piezoelectric film **19** is deposited on the first upper electrode **18**.

(43) At Step **10**, a first ground electrode **10** is deposited on the first piezoelectric film **19**. The material of the first ground electrode **10** is not limited herein.

Embodiment 2

(44) Referring to FIG. **7**, on the basis of the Embodiment 1, another implementation of a volatile threshold switching device TS is provided.

(45) The volatile threshold switching device TS includes: a second substrate **21**; a third isolation layer **22**, which is formed on the second substrate **21**; a second lower electrode **23**, which is formed on the third isolation layer **22**; a second functional layer **24**, which is formed on the third isolation layer **23**; and a second intermediate electrode **25**, which is formed on the second functional layer **24**; wherein a second resistive film **26** is deposited on the second intermediate electrode **25** to serve as the resistance Rc; and a second upper electrode **27** is deposited on the second resistive film **26**.

(46) In the present embodiment, the second substrate **21** is a silicon wafer; the third isolation layer **22** is a SiO.sub.2 layer; the material of the second lower electrode **23** is an inert conductive material, such as TiN, Poly-Si, Pd, Pt, W, or Au; the material of the second functional layer **24** is NbO.sub.2, VO.sub.2, SiTe, SiO.sub.2:Ag, a-Si:Cu, a-Si:Ag or AM.sub.4Q.sub.8; the material of the functional layer may, but be not limited to, be selected from the above materials, and any material having a volatile threshold switching characteristic can be used; the material of the second intermediate electrode **25** is a conductive material, such as TiN, Poly-Si, Pd, Pt, W, Cu, Ag, or Au; the resistance value of the second resistive film **26** is greater than 5 times of a resistance value of the second functional layer **24** in a low-resistance state and is less than ⅕ of a resistance value of the first functional layer **15** in a high-resistance state; and in this case, the AM.sub.4Q.sub.8 is a mixed material of Ga, Ge, V, Nb, Ta, Mo, S, and Se.

(47) Referring to FIG. **8**, the present disclosure further provides a method for preparing the volatile threshold switching device TS, which may include steps **1-6**.

(48) At Step **1**, a silicon wafer serves as a second substrate **21**, and oxidation is applied on the silicon wafer to form a SiO.sub.2 layer as a second isolation layer **22**. The thickness of the SiO.sub.2 layer is 100 nm˜300 nm. The thickness of the SiO.sub.2 layer may also be decreased or increased according to an actual process condition.

(49) At Step **2**, a second lower electrode **23** is deposited on the second isolation layer **22**. The thickness of the second lower electrode **23** is 10 nm˜100 nm.

(50) At Step **3**, a second functional layer **24** is deposited on the second lower electrode **23**. The thickness of the second functional layer **24** is 5 nm˜50 nm.

(51) At Step **4**, a second intermediate electrode **25** is deposited on the second functional layer **24**.

The thickness of the second intermediate electrode **25** is 10 nm˜100 nm.
(52) At Step **5**, a second resistive film **26** is deposited on the second intermediate electrode **25**. The resistance value of the second resistive film **26** is 1 Ω˜1 MΩ, which depends on a high-resistance or low-resistance state of the functional layer that is actually deposited.
(53) At Step **6**, a second upper electrode **27** is deposited on the second resistive film **26**. The material of the second upper electrode **27** is not limited herein.
(54) Referring to FIG. **9**, the present disclosure further provides a mechanoreceptive system, including: the aforementioned afferent neuron circuit; a second piezoelectric film **28**, which is formed on the second upper electrode **27**; and a second ground electrode **29**, which is deposited on the second piezoelectric film **28**.
(55) The material of the second piezoelectric film **28** may be BaTiO.sub.3/bacterial cellulose, K.sub.0.5Na.sub.0.5NbO.sub.3—BaTiO.sub.3/polyvinylidene fluoride, crystal, lithium gallate, lithium germanate, titanium germanate, or lithium tantalate.
(56) Referring to FIG. **10**, the present disclosure further provides a preparation method, which may include steps **7** and **8**.
(57) At Step **7**, on the basis of the aforementioned afferent neuron integration scheme, a second piezoelectric film **28** is deposited on the second upper electrode **27**. A piezoelectric material may, but be not limited to, be selected from the above materials, and any material having a piezoelectric characteristic can be used. However, a thermal budget of the functional layer should be taken into account during integration so as to determine the annealing temperature of the piezoelectric material.
(58) At Step **8**, a second ground electrode **29** is deposited on the second piezoelectric film **28**. The material of the second ground electrode **29** is not limited herein.
(59) The performance of the above circuits and systems is described below through graphs.
(60) Referring to FIG. **11**, a simulation graph of I-V characteristic curves is obtained by subjecting the volatile threshold switching devices TS prepared in the Embodiment 1 and Embodiment 2 to voltage scanning;
(61) When a voltage applied to the intermediate electrode exceeds a certain voltage value (V.sub.TH+ or V.sub.TH−), the volatile threshold switching device TS switches from a high-resistance state to a low-resistance state.
(62) In the process of voltage flyback, when the voltage is less than a certain voltage value (V.sub.HOLD+ or V.sub.HOLD−), the volatile threshold switching device TS returns from the low-resistance state to the high-resistance state as the voltage of the intermediate electrode is not adequate to maintain the low-resistance state of the volatile threshold switching device TS.
(63) Referring to FIG. **12**, shown is a simulation graph of output oscillation curves of the afferent neuron circuit subjected to two input voltages. The afferent neuron circuit is subjected to two input voltages to output oscillation curves. When an input is a constant voltage value, the afferent neuron outputs a stable oscillation signal, and the output frequency of the oscillation signal is related to the magnitude of the input voltage. This corresponds to a working mode of afferent neurons in an organism. The result indicates that the designed afferent neuron circuit can convert simulated input signals into different output oscillation frequencies, which can meet a requirement of an impulse neural network.
(64) Referring to FIG. **13**, shown is a simulation graph of the relationship between an output frequency and an input voltage of the afferent neuron circuits provided by the Embodiment 1 and Embodiment 2. Within a certain voltage range, an output frequency of the afferent neuron presents a linear increase relation relative to an input voltage. This indicates that the output frequency of the afferent neuron circuit can correctly reflect the strength of the stimulus input.
(65) Referring to FIG. **14**, shown is a simulation graph showing an output voltage curve of the piezoelectric film when pressure is applied thereon, an oscillation output curve, and an oscillation output frequency curve of the afferent neuron circuits provided by the Embodiment 1 and

Embodiment 2. The output frequency of the afferent neuron subjected to a large-amplitude sine wave input has two frequency peaks. Within a small range of the input voltage, the output frequency increases positively with the increase of the input voltage. When the input voltage reaches a certain value, the output frequency reaches the maximum saturation value. Then, the voltage is continually increased so that the output oscillation frequency gradually decreases until oscillation stops. This indicates that the afferent neuron circuit has a self-protection mechanism. That is, when an external stimulation signal is too strong in strength, the afferent neuron circuit can self-adjust the output frequency to adapt to the external environment, thereby maintaining subsequent stability of the system.

(66) It is ready to find that the circuit and system prepared according to the above embodiments all have good performances of an afferent neuron circuit.

(67) According to the embodiments of the present disclosure, one or more technical solutions have at least the following technical effects or advantages.

(68) According to the afferent neuron circuit and the mechanoreceptive system that the embodiments of the present disclosure have provided, an afferent neuron circuit is formed with a simple structure by connecting a resistance Rc and a volatile threshold switching device TS in series. A connection point between the volatile threshold switching device TS and the resistance Rc is used as a signal output terminal. An input signal is a voltage signal or a current signal, and an output signal is an oscillation frequency signal. The oscillation frequency of the output signal is related to the strength of the input signal. The volatile threshold switching device TS is provided with a parasitic capacitor C.sub.parasitic, so that when there is an input signal at the signal input terminal, the TS device is charged by the circuit through the resistance Rc; when a voltage across the parasitic capacitor Cparasitic exceeds a switching voltage of the TS device, the TS device switches to a low-resistance state, and the capacitor discharges through the TS device; when the voltage across the parasitic capacitor Cparasitic drops to a holding voltage of the TS device, the TS device switches to a high-resistance state, and the parasitic capacitor Cparasitic is charged again through the resistance Rc. These are repeated over and over again to form an oscillation output pulse signal, thereby achieving functional integrity.

(69) The above are only preferred embodiments of the present disclosure and are not used to limit the protection scope of the present disclosure. Any modification, equivalent replacement, and improvement made within the spirit and principle of the present disclosure should be included in the protection scope of the present disclosure.

## Claims

1. An afferent neuron circuit, comprising a resistance (Rc) and a volatile threshold switching device (TS); wherein the volatile threshold switching device (TS) has a parasitic capacitor (Cparasitic); a first end of the resistance (Rc) serves as a signal input terminal, and a second end of the resistance (Rc) serves as a signal output terminal; a first end of the volatile threshold switching device (TS) is connected to the signal output terminal, and a second end of the volatile threshold switching device (TS) is grounded, wherein the afferent neuron circuit has an integrated semiconductor structure, wherein the volatile threshold switching device (TS) comprises: a first substrate; a first isolation layer formed on the first substrate; a first lower electrode formed on the first isolation layer; a second isolation layer formed on the first lower electrode; a first functional layer formed on the second isolation layer wherein a middle part of the first functional layer is connected to the first lower electrode through an etched via hole; and a first intermediate electrode formed on the first functional layer; wherein a first resistive film is deposited on the first intermediate electrode to serve as the resistance (Rc), and a first upper electrode is deposited on the first resistive film.
2. The afferent neuron circuit according to claim 1, wherein, the first substrate is a silicon wafer; the first isolation layer is a $SiO_2$ layer; a material of the first lower electrode is TiN, Poly-Si, Pd, Pt,

W, or Au; the second isolation layer is a $SiO_2$ layer; a material of the first functional layer is $NbO_2$, $VO_2$, SiTe, $SiO_2$:Ag, a-Si:Cu, a-Si:Ag, or $AM_4Q_8$; a material of the first intermediate electrode is TiN, Poly-Si, Pd, Pt, W, Cu, Ag or Au; a resistance value of the first resistive film is greater than 5 times of a resistance value of the first functional layer in a low-resistance state and is less than ⅕ of a resistance value of the first functional layer in a high-resistance state; wherein the $AM_4Q_8$ is a mixed material of Ga, Ge, V, Nb, Ta, Mo, S, and Se.

3. A mechanoreceptive system, comprising: the afferent neuron circuit according to claim 1; a first piezoelectric film formed on the first upper electrode of the afferent neuron circuit; and a first ground electrode deposited on the first piezoelectric film.

4. The mechanoreceptive system according to claim 3, wherein a material of the first piezoelectric film is $BaTiO_3$/bacterial cellulose, $K_{0.5}Na_{0.5}NbO_3$-$BaTiO_3$/polyvinylidene fluoride, crystal, lithium gallate, lithium germanate, titanium germanate, or lithium tantalite.

5. An afferent neuron circuit, comprising a resistance (Rc) and a volatile threshold switching device (TS); wherein the volatile threshold switching device (TS) has a parasitic capacitor (Cparasitic); a first end of the resistance (Rc) serves as a signal input terminal, and a second end of the resistance (Rc) serves as a signal output terminal; a first end of the volatile threshold switching device (TS) is connected to the signal output terminal, and a second end of the volatile threshold switching device (TS) is grounded, wherein the afferent neuron circuit has an integrated semiconductor structure, wherein the volatile threshold switching device (TS) comprises: a first substrate; a first isolation layer formed on the first substrate; a first lower electrode formed on the first isolation layer; a second isolation layer formed on the first lower electrode; a first functional layer formed on the second isolation layer wherein a middle part of the first functional layer is connected to the first lower electrode through an etched via hole; and a first intermediate electrode formed on the first functional layer.

6. The afferent neuron circuit according to claim 5, wherein the second substrate is a silicon wafer; the third isolation layer is a $SiO_2$ layer; a material of the second lower electrode is TiN, Poly-Si, Pd, Pt, W, or Au; a material of the second functional layer is $NbO_2$, $VO_2$, SiTe, $SiO_2$:Ag, a-Si:Cu, a-Si:Ag, or $AM_4Q_8$; a material of the second intermediate electrode is TiN, Poly-Si, Pd, Pt, W, Cu, Ag or Au; a resistance value of the second resistive film is greater than 5 times of a resistance value of the second functional layer in a low-resistance state and is less than ⅕ of a resistance value of the first functional layer in a high-resistance state; wherein the $AM_4Q_8$ is a mixed material of Ga, Ge, V, Nb, Ta, Mo, S, and Se.

7. A mechanoreceptive system, comprising: the afferent neuron circuit according to claim 5; a second piezoelectric film formed on the second upper electrode of the afferent neuron circuit; and a second ground electrode deposited on the second piezoelectric film.

8. The mechanoreceptive system according to claim 7, wherein a material of the second piezoelectric film is $BaTiO_3$/bacterial cellulose, $K_{0.5}Na_{0.5}NbO_3$-$BaTiO_3$/polyvinylidene fluoride, crystal, lithium gallate, lithium germanate, titanium germanate, or lithium tantalate.